Generalization of $3n + 1$ Collatz Orbits *via* Jacobsthal Sequences:

Visualization using Voronoi Tessellation in 2-adic Logarithmic Spiral Space

Hiroshi Harada - May 23, 2026

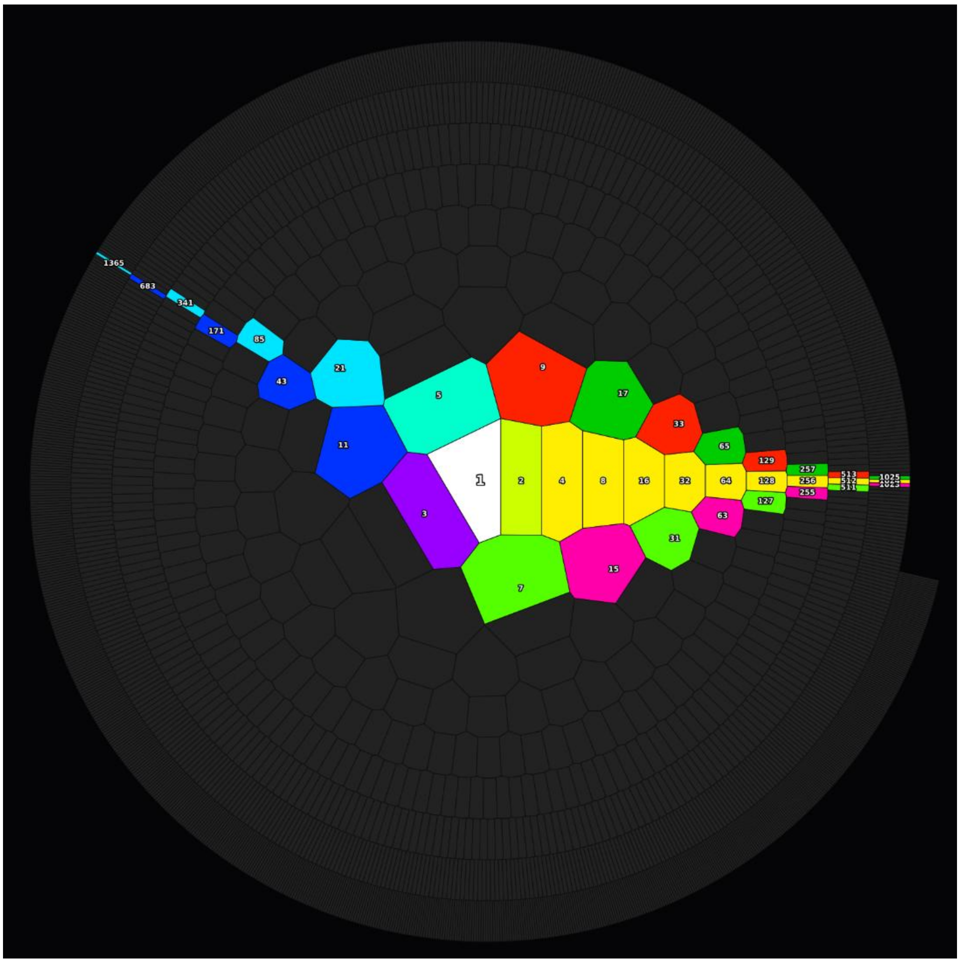

Figure 1. The Stained-Glass Topology of the Collatz Map.
Visualization of the natural number space around the final attractor $1 \cdot 2^b$. Mapped onto a 2-adic logarithmic spiral and partitioned *via* Voronoi tessellation, the space reveals a highly ordered geometric structure governed by seven Jacobsthal-type sequences. The cyan trajectory illustrates the final $5 \to 1$ transition of the odd Collatz orbit, landing seamlessly onto the $1 \cdot 2^b$ axis.